

Anshuman Pandey

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PROFESSIONAL SUMMARY

CSE (AI/ML) student at VIT Bhopal with hands-on experience across the machine learning lifecycle, from model development to production deployment and monitoring. Built deep learning and computer vision systems achieving up to 98.2% accuracy, including a chess engine using MCTS and reinforcement learning, and a medical imaging classifier fine-tuned on retinal fundus data. Also developed an end-to-end MLOps pipeline covering feature engineering, model serving via FastAPI, and automated drift monitoring with CI/CD. Proficient in Python, PyTorch, TensorFlow, FastAPI, and Docker. Actively seeking internship or entry-level roles in Machine Learning Engineering or Data Science.

EXPERIENCE

ServiceNow Virtual Intern | May 2026

Completed a virtual internship program focused on ServiceNow platform administration and automation. Gained practical knowledge of ServiceNow Administration, Flow Designer, Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Vellore Institute of Technology, Bhopal

2023 – Present | CGPA: 8.52/10

Class XII: 89.5% | Class X: 89.4%

PROJECTS

AlphaZ0 — Neural Chess Engine | Python, PyTorch, MCTS, A3C, Pygame | 2026

- Built a fully playable chess engine from scratch with complete rule support (castling, en passant, promotion, legal move filtering) and an 18-plane board encoder for neural input.
- Designed a ResNet dual-head neural network (10 residual blocks, 128 channels) with separate policy and value heads, modeled after DeepMind's AlphaZero architecture.
- Implemented MCTS with UCB selection and Dirichlet noise at root, combined with a parallel A3C training loop where worker threads compute GAE-advantage gradients asynchronously.
- Built a Pygame UI with main menu, algebraic notation moves log sidebar, bot-vs-human mode with color selection, promotion dialog, and board flip for black.

<https://github.com/AnshumanJ28/AlphaZ0>

MLOps Demand Forecasting Pipeline | Python, LightGBM, FastAPI, DVC, MLflow, Evidently AI, Docker | 2026

- Built an end-to-end MLOps pipeline for spatio-temporal traffic demand forecasting, covering data versioning, feature engineering, model training, REST API serving, and automated drift monitoring.
- Engineered temporal and cyclical features (geohash encoding, rush-hour flags, sine/cosine hour encoding) and trained a LightGBM regression model, logging RMSE, MAE, and R^2 via MLflow tracked on DagsHub.
- Deployed the model as a FastAPI/Uvicorn REST endpoint for real-time demand prediction, containerized with Docker.
- Implemented Evidently AI drift monitoring and a GitHub Actions CI/CD pipeline running automated unit tests on every push to main.
- Versioned datasets and pipeline artifacts with DVC and a Google Drive remote.

<https://github.com/AnshumanJ28/mlops-demand-forecasting>

Diabetic Retinopathy Detection | Python, PyTorch, EfficientNet-B5, timm | 2025

- Fine-tuned a pretrained EfficientNet-B5 on the APTOS 2019 and RFMiD retinal fundus image datasets for binary classification of diabetic retinopathy risk.
- Applied two-phase transfer learning — classifier-head-only training followed by full fine-tuning at reduced LR — using AdamW with cosine annealing scheduling.
- Handled class imbalance via weighted BCE loss and augmentation (flips, rotation, colour jitter); evaluated with accuracy and MCC for robust performance on imbalanced medical data.

<https://github.com/AnshumanJ28/Diabatic-B5>

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, SQL (MySQL), HTML, CSS
- **ML / AI:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, CNN, LSTM, FFT preprocessing
- **MLOps & Deployment:** LightGBM, FastAPI, Docker, DVC, MLflow, GitHub Actions (CI/CD), Evidently AI
- **Tools & Platforms:** AWS, Git, Jupyter Notebook, Google Colab, Kaggle Notebook, Advanced Excel, Canva, N8N
- **Concepts:** Data Structures & Algorithms, Computer Vision, Front-end Development

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank) | **Python & Java** (HackerRank) | **Cloud Computing** (NPTEL) | **Advanced Excel** (Udemy) | **AI & Agentic AI** (TalentSprint)|**ML** (Coursera)| **AWS trainer**(AWS)|**AWS Academy**(AWS)|**IoT**(NPTEL)| **Service Now**

ACHIEVEMENTS

- 2nd Runner-up, IEEE-NSUT Ideathon — awarded for ML-based project among 80+ competing teams.
- Top 10 finish, BITS Pilani Hackathon — among top teams nationally in a competitive problem-solving event.